

PROVISIONAL PATENT APPLICATION

United States Patent and Trademark Office

SEGMENTED TUBULAR THREE-DIMENSIONAL OPTOELECTRONIC PACKAGE SUBSTRATE WITH DIRECTIONAL MULTI-TUBE OPTICAL CHANNELS AND VERTICAL CHIPLET INTEGRATION

Inventor	River Chen
Filing Type	Provisional Patent Application (35 U.S.C. § 111(b))
Draft Date	June 2026
Docket Reference	GCF-ASL-TUBULAR-PACKAGE-002
Document Status	Pre-PoC architecture disclosure draft

CONFIDENTIAL - UNFILED DRAFT

Performance values are illustrative design estimates unless expressly identified as measured results.

1. TITLE OF THE INVENTION

SEGMENTED TUBULAR THREE-DIMENSIONAL OPTOELECTRONIC PACKAGE SUBSTRATE WITH DIRECTIONAL MULTI-TUBE OPTICAL CHANNELS AND VERTICAL CHIPLET INTEGRATION

2. FIELD OF THE INVENTION

The present disclosure relates to semiconductor packaging, heterogeneous integration, chiplet interconnection, package substrates, optical interconnects, and three-dimensional electronic assemblies. More particularly, it relates to a hollow tubular package-substrate structure formed from multiple arcuate substrate segments, each providing one or more optical, electrical, mechanical, thermal, and chiplet-interface functions. Multiple tubular package substrates may be arranged on a common carrier and assigned predetermined axial optical propagation directions, including top-to-bottom and bottom-to-top directions, in symmetric or asymmetric ratios.

3. BACKGROUND OF THE INVENTION

Advanced processors increasingly combine compute dies, memory stacks, input/output dies, accelerators, and control dies as chiplets within a common package. Existing two-dimensional and two-and-one-half-dimensional integration schemes generally place chiplets on a substantially planar interposer, redistribution structure, bridge, or package substrate. These architectures provide high-density interconnection but remain constrained by planar footprint, long electrical routing paths, power-delivery congestion, thermal concentration, reticle and interposer scaling limits, and the increasing energy cost of high-speed copper communication.

Optical interconnect technologies can reduce the electrical distance traveled by high-bandwidth signals, but many implementations treat optical components as separate modules added to an otherwise planar package. Such arrangements do not fully exploit the third dimension as an active package-substrate surface and may require long lateral redistribution paths before signals reach the intended chiplets.

There is therefore a need for a package architecture in which the package substrate itself is converted from a primarily planar structure into a segmented tubular three-dimensional structure capable of supporting chiplets around its outer surface, carrying optical signals along its axial extent, and concentrating selected copper conductors in seam regions between adjacent substrate segments. There is also a need for multi-tube systems in which different tubes provide different unidirectional optical-flow roles according to the traffic requirements of a particular workload.

4. SUMMARY OF THE INVENTION

The invention provides a segmented tubular optoelectronic package substrate. The tubular substrate is assembled from a plurality of arcuate package-substrate segments positioned around a hollow interior. Each arcuate segment may include dielectric build-up layers, redistribution conductors, optical coupling structures, photodetectors, optical modulators, optical transmitters, optical receivers, through-substrate conductors, chiplet bond pads, power-delivery structures, and thermal-management features.

Adjacent arcuate segments are joined at seam regions. One or more conductive seam interconnect structures are positioned at or near the seam regions and may provide electrical power, ground, bias, control, calibration, management signaling, low-speed communication, fault bypass, mechanical retention, or thermal conduction. By concentrating selected copper conductors at the segment boundaries, the architecture spatially separates the principal axial optical channels from at least part of the electrical distribution network.

The hollow interior and/or the inner region of the tubular package substrate includes at least one optical propagation structure extending generally along the tube axis. A given tube preferably has one predominant axial optical-flow direction: either top-to-bottom or bottom-to-top. Multiple tubes may be

arranged on a common package carrier. For a four-tube embodiment, the directional allocation may be two downlink tubes and two uplink tubes, three downlink tubes and one uplink tube, one downlink tube and three uplink tubes, or another allocation selected according to bandwidth, latency, redundancy, and workload requirements.

Unlike a passive optical liner, the tubular body is itself a package-substrate variant. It provides a three-dimensional chiplet attachment surface, local optical-electrical conversion, power and control distribution, mechanical support, and selected signal routing. The disclosed architecture may coexist with conventional planar package substrates, redistribution layers, local silicon bridges, glass interposers, organic interposers, or hybrid-bonded structures.

5. DEFINITIONS AND ARCHITECTURAL PRINCIPLES

5.1 Tubular Package Substrate

A “tubular package substrate” is a hollow or partially hollow package-support structure having an axial length and a closed or partially closed circumferential wall. The wall may be circular, elliptical, polygonal with arcuate portions, or another generally tubular geometry. The term does not require a perfect mathematical cylinder.

5.2 Arcuate Substrate Segment

An “arcuate substrate segment” is a curved package-substrate panel forming part of the tubular wall. It may be fabricated in a substantially planar state and subsequently curved, or fabricated directly with curvature. The segment may comprise semiconductor, glass, ceramic, organic laminate, polymer, silicon, silicon nitride, sapphire, composite material, or combinations thereof.

5.3 Predominant Axial Optical Direction

A “predominant axial optical direction” means that a tube is configured primarily for optical energy or optically encoded information to propagate from a first axial end toward a second axial end. Local radial coupling between the axial optical path and devices on the tube wall does not change the predominant direction. Bidirectional auxiliary signaling may be present without converting the tube into a symmetric bidirectional channel.

5.4 Seam Interconnect Structure

A “seam interconnect structure” is an electrically conductive or hybrid structural element located between, across, or adjacent to neighboring arcuate segments. It may include copper, copper alloy, tungsten, aluminum, conductive adhesive, flexible printed circuitry, redistribution traces, vertical rails, metalized ribs, local bridges, or combinations thereof.

6. OVERALL SYSTEM ARCHITECTURE

6.1 Single-Tube Assembly

A single tubular assembly includes a plurality of arcuate substrate segments arranged around a hollow axial region. In one non-limiting embodiment, three segments each have a nominal angular span near 120 degrees. In other embodiments, two, four, five, six, or more segments may be used. Segment edges may include non-functional keep-out regions, bonding lands, alignment features, sealing structures, or conductive seam strips.

The inner-facing portion of each segment may include optical coupling elements communicating with one or more axial optical propagation structures. The outer-facing portion includes chiplet attachment zones. Chiplets may be attached directly to the curved substrate, to locally planar terraces, to stepped mounting lands, to intermediate bridge elements, or to flexible or compliant interconnect regions.

6.2 Multi-Tube Assembly

A plurality of tubular package substrates may be mounted on a common lower package carrier, an upper optical distribution member, or both. The tubes may be identical or heterogeneous in diameter, height, number of segments, optical wavelength, optical-flow direction, chiplet population, electrical seam configuration, cooling structure, or function.

In a four-tube implementation, the system controller may allocate the tubes according to traffic asymmetry. A compute-intensive system that receives substantially more model, weight, activation, and command data than it returns as final output may use three top-to-bottom tubes and one bottom-to-top tube. A balanced communication system may use two tubes in each direction. A sensing or data-collection system may use one top-to-bottom control tube and three bottom-to-top collection tubes.

6.3 Example Directional Allocations

Configuration	Top-to-Bottom	Bottom-to-Top	Illustrative Use	Notes
2:2	2	2	Balanced data exchange	Symmetric capacity and redundancy
3:1	3	1	AI/HPC input-dominant workload	Separate command, weight/data, and activation downlinks with aggregated uplink
1:3	1	3	Sensor or distributed-result collection	Shared command downlink with multiple result uplinks
4:0 or 0:4	4 or 0	0 or 4	Specialized broadcast or collection stage	May be used as one stage within a larger package

7. ARCUATE SEGMENT STRUCTURE

7.1 Substrate Layers

Each arcuate segment may include one or more dielectric build-up layers, redistribution layers, conductive vias, embedded passives, power and ground conductors, optical waveguide layers, optical coupling regions, active photonic devices, thermal-spreading layers, and external bond pads. The segment may be fabricated as an organic package-substrate segment, a glass segment, a ceramic segment, a semiconductor segment, or a heterogeneous stack.

7.2 Inner Optical Interface

The inner side of a segment may interact with an axial optical path by means of apertures, grating couplers, prism couplers, reflective facets, evanescent couplers, lens structures, wavelength-selective couplers, photodetector arrays, modulators, optical emitters, optical transceivers, or combinations thereof. Optical components may be integrated into the substrate layers or attached as localized photonic tiles.

7.3 Outer Chiplet Interface

The outer side of each segment provides chiplet attachment zones. The attachment zones may include micro-bumps, copper pillars, hybrid-bond surfaces, anisotropic conductive material, compression contacts, flexible connectors, socket structures, or direct bonded interfaces. Local planar mounting terraces may be formed on an otherwise curved segment to simplify chiplet attachment.

7.4 Local Electrical Conversion

A received optical signal may be converted into an electrical signal by a photodetector and supplied to one or more attached chiplets. Conversely, an electrical output produced by an attached chiplet may drive a modulator or emitter that injects an outbound optical signal into the axial optical channel of an uplink tube. The light source may be integrated locally, attached to a segment, positioned at an axial end, or supplied remotely as a continuous-wave optical carrier.

8. SEAM-LOCATED COPPER AND ELECTRICAL INTERCONNECT

8.1 Seam Electrical Spine

Selected copper conductors are preferably concentrated at the boundaries between adjacent arcuate segments. A seam may include a vertical conductive spine extending along part or all of the tube height. The spine may connect power, ground, detector bias, modulator bias, management signals, calibration signals, fault-status signals, and other electrical functions among the segments and between the tube and a planar carrier.

8.2 Spatial Separation from the Optical Region

The conductive seam structures may be located radially outward of the principal optical propagation region and separated from that region by dielectric, optical isolation, shielding, reflective boundaries, absorber structures, or physical spacing. This reduces undesired optical absorption, back-reflection, electrical noise coupling, and receiver interference.

8.3 Seam Strip Embodiment

In one embodiment, a prefabricated seam strip is positioned between adjacent substrate segments. The seam strip may be a flexible copper-polyimide circuit, narrow redistribution bridge, metalized structural rib, glass interconnect strip, semiconductor bridge, or composite connector. It may simultaneously provide electrical connection, segment alignment, mechanical retention, thermal conduction, and sealing.

8.4 Division of Electrical and Optical Functions

High-bandwidth or long-distance data transport may be assigned primarily to optical channels, while copper seam conductors carry power, ground, management, calibration, low-speed control, or short local electrical signals. The invention does not require complete elimination of copper interconnects and may use hybrid optical-electrical communication according to signal type and distance.

9. OPTICAL PROPAGATION AND DIRECTIONAL TUBE OPERATION

9.1 Top-to-Bottom Tube

A top-to-bottom tube receives optical energy or optically encoded data at or near its upper axial end. The signal propagates generally downward and may be coupled to devices positioned at one or more axial levels. Different downlink tubes may separately carry clock and control, model or weight data, input activations, configuration information, or other signal classes.

9.2 Bottom-to-Top Tube

A bottom-to-top tube receives optical signals generated at lower axial levels or by chiplets attached to its wall and transports the signals generally upward. Multiple local transmitters may be combined through wavelength-division multiplexing, time-division multiplexing, spatial multiplexing, code-division multiplexing, arbitration, or optical combining structures.

9.3 Optical Path Implementations

The axial optical propagation structure may comprise a hollow reflective passage, one or more wall-integrated waveguides, embedded optical fibers, dielectric waveguides, photonic-crystal structures, free-space optical channels, segmented waveguide rails, or hybrid arrangements. The invention is not limited to a multi-bounce reflective cavity.

9.4 Delay and Skew Management

Rotational symmetry can provide substantially equal path conditions for receiving elements located at comparable axial positions. Receivers at different axial levels may use separate source positions, segmented channels, predetermined waveguide lengths, refractive-index compensation, adjustable delay elements, electronic deskew, wavelength-specific routing, or calibrated phase correction. Accordingly, the disclosure does not rely on an assumption that every point on an extended cylindrical surface is automatically equidistant from a single point source.

10. EXAMPLE CHIP-PROCESSING LOGIC

The physical direction of each optical tube is distinct from the logical function of any individual chiplet. A tube may carry one or more classes of signals, and chiplets attached to different tubes may cooperate through the lower carrier, upper optical layer, local bridges, or direct inter-tube links.

10.1 Illustrative 3:1 AI/HPC Configuration

- Downlink Tube 1: clock, synchronization, control, addressing, and task-dispatch information.
- Downlink Tube 2: model weights, memory blocks, or shared read-only data.
- Downlink Tube 3: input data, activations, or intermediate data returned to a later compute stage.
- Uplink Tube 4: processed results, telemetry, error reports, acknowledgments, and external output.

Compute and memory chiplets may be distributed around one or more tubes. Local electrical paths may be used for very short connections, while optical channels transport selected signals along the tube height or between package levels. The workload may be pipelined so that different axial levels process different stages concurrently.

10.2 Illustrative 2:2 Balanced Configuration

- Two downlink tubes carry data and control classes independently or redundantly.
- Two uplink tubes return results and status or provide independent output fabrics.
- A failed or degraded tube may be bypassed by reassigning channels, wavelengths, or functions to another tube.

11. FABRICATION AND ASSEMBLY METHODS

11.1 Planar Fabrication Followed by Curving

1. Fabricate redistribution layers, electrical vias, optical coupling structures, and chiplet interfaces on a substantially planar substrate panel.
2. Define segment edges, seam keep-out regions, alignment marks, and optional local mounting terraces.
3. Thin or otherwise prepare the panel when curvature requires reduced bending stiffness.
4. Separate the panel into arcuate-segment preforms.
5. Curve each segment using mechanical forming, thermoforming, elastic bending, lamination to a curved mandrel, or another controlled forming process.
6. Join multiple segments to form a tubular package substrate and install seam interconnect strips or conductive bridges.
7. Attach photonic tiles, chiplets, heat spreaders, optical end structures, and carrier connections as appropriate.

11.2 Direct Curved Fabrication

In alternative embodiments, segments are fabricated directly from molded glass, ceramic, organic laminate, additive-manufactured dielectric, curved semiconductor material, or another pre-curved body. Conductors and optical structures may then be deposited, printed, plated, bonded, embedded, or attached.

11.3 Modular Replacement

Because the tubular wall is segmented, an individual defective segment may be tested before final assembly and, in some embodiments, replaced without discarding all other segments. The seam structures may provide detachable, reworkable, or permanently bonded connections.

12. THERMAL AND MECHANICAL MANAGEMENT

The tubular geometry provides a large outer surface for chiplet placement and heat-transfer structures relative to its planar footprint. Cooling structures may include axial airflow through the hollow interior, external fins, liquid-cooling jackets, heat pipes, vapor chambers, thermally conductive seam ribs, top and bottom heat spreaders, or combinations thereof.

Mechanical stress may be managed through segment thickness selection, neutral-axis placement, compliant dielectric layers, segmented curvature, strain-relief features, low-modulus adhesives, local rigid islands, expansion gaps, and materials selected according to coefficient of thermal expansion. No specific bend radius, fracture limit, or thermal performance is required by the broad architecture.

12.1 Distinction Between Heat Generation and Heat-Flux Density

The tubular geometry does not by itself eliminate heat produced by computation, memory access, optical conversion, or power delivery. Its direct geometric effect is to increase usable external surface area and to redistribute heat-transfer paths. Separately, replacing selected long-distance electrical interconnects with optical channels may reduce interconnect, repeater, serialization, and global-distribution power. These two effects are distinct and may be combined in a system-level thermal design.

12.2 Conservative Thermal-Improvement Basis

For an equivalent planar footprint, the cylindrical or partially cylindrical sidewalls may provide more than twice the externally accessible heat-transfer area of a planar package, depending on tube height, radius, spacing, chiplet coverage, and cooling access. Under comparable total power, an effective heat-transfer area exceeding twice the planar reference area corresponds to an average heat-flux density below one-half of the planar reference value. This statement is a geometric design estimate rather than a measured thermal result.

Where optical channels also reduce electrical interconnect and global clock-distribution losses, the reduction in generated heat may compound the area-based reduction in heat-flux density. The architecture is therefore intended to reduce thermal concentration by more than a factor of two in suitable implementations, while recognizing that the actual result depends on workload, chiplet power, optical-device efficiency, contact thermal resistance, airflow, heat spreaders, and cooling hardware.

12.3 Common Thermal Platform Rather Than Multiple Heat-Driven Package Topologies

A further objective is to reduce the need for multiple fundamentally different package topologies created primarily to avoid thermal concentration. The same tubular substrate platform may support different power classes by varying tube height, diameter, chiplet population, chiplet spacing, downlink-to-uplink tube ratio, optical-channel count, seam-conductor capacity, and attached cooling structures. Thermal management remains necessary, but may be implemented as a configurable parameter of a common platform rather than requiring a separate base architecture for each thermal class.

12.4 Local Thermal Controls

Local thermal controls may include chiplet-side heat spreaders, axial airflow through or around the hollow region, external fins, liquid or two-phase cooling attachments, thermally conductive seam members, distributed temperature sensors, workload migration, and spacing between neighboring tubes. Such controls may be applied selectively without changing the fundamental tubular package-substrate architecture.

13. INTEGRATION WITH EXISTING PACKAGE TECHNOLOGIES

The tubular package substrate may be mounted on or integrated with an organic package substrate, redistribution interposer, glass interposer, silicon interposer, local silicon bridge, embedded bridge, fan-out structure, hybrid-bonded stack, memory stack, or other package platform. The tubular structure may supplement rather than replace planar high-density interconnection.

For example, a lower planar carrier may provide global power delivery and communication among multiple tubes, while each tube provides a three-dimensional chiplet surface and axial optical channels. Local high-density semiconductor bridges may be used only where needed, while larger mechanical and routing regions are implemented using lower-cost substrate materials.

14. ILLUSTRATIVE DESIGN ANALYSIS AND NON-LIMITING TARGETS

The tubular architecture can increase the available chiplet attachment surface per unit planar footprint because usable sidewall area increases with tube height. For an idealized circular tube, sidewall area scales approximately with $2\pi RH$, while footprint scales approximately with πR^2 . Practical usable area depends on tube spacing, seam width, keep-out zones, cooling features, chiplet dimensions, attachment terraces, and carrier layout.

Optical channels may reduce energy consumed by selected long-distance electrical links, but actual energy per bit, optical loss, skew, thermal flux, and system power depend on device technology, wavelength, modulation format, detector efficiency, driver circuitry, workload, packaging losses, and cooling design. Any numerical values developed before electromagnetic simulation, thermal analysis, stress analysis, and prototype measurement are design estimates rather than guaranteed performance.

15. ADVANTAGES AND DISTINGUISHING FEATURES

- The package substrate itself is converted into a three-dimensional tubular chiplet-support structure rather than merely receiving a separate optical tube.
- Arcuate segments allow planar or near-planar fabrication before three-dimensional assembly.
- The hollow axial region supports top-to-bottom or bottom-to-top optical channels.
- Multiple tubes can be assigned different unidirectional optical-flow directions in workload-dependent ratios.
- Copper conductors can be concentrated at segment seams as vertical electrical spines separated from the principal optical region.
- Localized photonic and semiconductor regions may be combined with lower-cost passive substrate regions.
- The architecture supports direct chiplet attachment around a vertical surface and can coexist with planar interposers, bridges, and package substrates.
- Segment-level testing, replacement, redundancy, and heterogeneous tube designs may improve system configurability and yield management.

The tubular sidewall geometry can reduce average heat-flux density by increasing externally accessible heat-transfer area for a given planar footprint.

A common tubular platform may reduce product-line fragmentation caused by thermal constraints, because different power classes can be implemented through dimensional, population, channel, and cooling-attachment changes rather than entirely different package topologies.

16. CLAIMS FOR DISCLOSURE SUPPORT

The following claims are included to organize contemplated scope. They are non-final and may be revised in a later non-provisional application.

1. A three-dimensional optoelectronic package assembly comprising: a tubular package substrate extending along an axis and surrounding a hollow interior, the tubular package substrate comprising a plurality of arcuate substrate segments joined at respective seam regions; at least one optical propagation structure extending generally along the axis and configured to carry an optical signal in a predominant axial direction; a plurality of chiplet attachment zones disposed on outer-facing portions of the arcuate substrate segments; and at least one electrically conductive seam interconnect structure positioned at or adjacent to a seam region between neighboring arcuate substrate segments.
2. The assembly of claim 1, wherein the electrically conductive seam interconnect structure comprises a conductor extending along at least part of the axial length and is configured to provide one or more of power distribution, grounding, bias, management signaling, calibration signaling, fault signaling, inter-segment communication, mechanical retention, or thermal conduction.
3. The assembly of claim 1, wherein at least one arcuate substrate segment comprises a dielectric build-up layer, a redistribution conductor, an optical coupling element, an optical receiver, an optical transmitter, an optical modulator, a through-substrate conductor, or a combination thereof.
4. The assembly of claim 1, wherein the chiplet attachment zones are located on locally planar terraces formed on an otherwise curved outer surface.
5. The assembly of claim 1, wherein the predominant axial direction is a top-to-bottom direction and the optical propagation structure supplies optical signals to devices at a plurality of axial positions.
6. The assembly of claim 1, wherein the predominant axial direction is a bottom-to-top direction and optical signals generated at a plurality of axial positions are transported toward an upper axial end.
7. The assembly of claim 1, wherein the optical propagation structure comprises a hollow reflective passage, a wall-integrated dielectric waveguide, an embedded optical fiber, a free-space channel, a photonic-crystal channel, or a combination thereof.
8. The assembly of claim 1, wherein the electrically conductive seam interconnect structure is radially separated from the optical propagation structure by a dielectric isolation region, optical isolation region, shielding region, or physical spacing.
9. A multi-tube semiconductor package comprising a plurality of assemblies according to claim 1 mounted on a common carrier, wherein a first subset of the tubular package substrates is configured for predominant top-to-bottom optical propagation and a second subset is configured for predominant bottom-to-top optical propagation.
10. The package of claim 9, wherein the plurality of tubular package substrates consists of four tubular package substrates allocated in a ratio selected from two top-to-bottom to two bottom-to-top, three top-to-bottom to one bottom-to-top, or one top-to-bottom to three bottom-to-top.
11. The package of claim 9, wherein different tubular package substrates differ in at least one of diameter, height, segment count, wavelength allocation, optical source position, optical component population, chiplet population, seam conductor configuration, cooling configuration, or communication function.

12. The package of claim 9, further comprising a planar package substrate, redistribution layer, glass interposer, silicon interposer, local semiconductor bridge, embedded bridge, or hybrid-bonded structure electrically or optically coupled to at least one tubular package substrate.
13. The package of claim 9, wherein three tubular package substrates provide top-to-bottom channels respectively assigned to control or synchronization information, model or memory data, and input or activation data, and a fourth tubular package substrate provides a bottom-to-top channel for processed results or status information.
14. A method of fabricating a tubular optoelectronic package substrate, comprising: forming one or more electrical redistribution structures and one or more optical interface structures on a substantially planar substrate panel; defining a plurality of segment preforms; curving the segment preforms; arranging the curved segment preforms around a hollow interior; joining neighboring segment preforms at seam regions; and installing at least one conductive seam interconnect structure at or adjacent to one of the seam regions.
15. The method of claim 14, further comprising testing individual segment preforms before joining the segment preforms to form the tubular package substrate.
16. The method of claim 14, further comprising attaching one or more chiplets to outer-facing chiplet attachment zones and coupling the chiplets to an optical receiver, optical transmitter, optical modulator, or optical transceiver associated with the tubular package substrate.
17. A method of operating a multi-tube semiconductor package, comprising: assigning a first number of tubular package substrates to top-to-bottom optical channels; assigning a second number of tubular package substrates to bottom-to-top optical channels; communicating data to chiplets through the top-to-bottom optical channels; processing the data using the chiplets; and communicating processed information through the bottom-to-top optical channels, wherein the first number and second number are selected according to an expected directional traffic ratio.
18. The method of claim 17, further comprising dynamically reassigning a wavelength, signal class, or tube function in response to workload, congestion, temperature, fault status, or channel degradation.
19. The assembly of claim 1, wherein the tubular package substrate provides an externally accessible heat-transfer surface area exceeding twice a planar footprint area occupied by the tubular package substrate.
20. The package of claim 9, wherein a common tubular package-substrate architecture is configured for a plurality of system power classes by varying at least one of tube height, tube diameter, chiplet population, chiplet spacing, directional tube allocation, optical-channel count, seam-conductor capacity, or an attached cooling structure without changing the fundamental tubular package-substrate topology.

17. ABSTRACT

A segmented tubular three-dimensional optoelectronic package substrate is formed from a plurality of arcuate package-substrate segments surrounding a hollow axial region. The tubular body is itself a package-substrate variant and provides chiplet attachment zones on its outer surface, one or more axial optical propagation structures, local optical-electrical interface elements, and electrical distribution features. Conductive seam interconnect structures are positioned between adjacent arcuate segments and may form vertical copper spines for power, ground, bias, management, calibration, low-speed communication, mechanical retention, or thermal conduction while remaining spatially separated from the principal optical region. Multiple tubular package substrates may be mounted on a common carrier and assigned predominant unidirectional optical-flow directions. In a four-tube embodiment, top-to-bottom and bottom-to-top tubes may be allocated in 2:2, 3:1, or 1:3 ratios according to workload traffic. The

architecture supports heterogeneous chiplet integration around vertical surfaces and may coexist with planar package substrates, redistribution interposers, local semiconductor bridges, glass structures, or hybrid-bonded assemblies.

18. BRIEF DESCRIPTION OF DRAWINGS

Figure	Description
FIG. 1	Exploded perspective view of one tubular package substrate showing arcuate segments, seam strips, hollow axial region, and external chiplet zones.
FIG. 2	Radial cross-section showing inner optical propagation structure, substrate wall layers, local optical interface, outer chiplet, and seam-located copper spine.
FIG. 3	Axial cross-section of a top-to-bottom tube with optical input at the upper axial end and extraction at multiple axial levels.
FIG. 4	Axial cross-section of a bottom-to-top tube with local optical transmitters and aggregated optical output at the upper axial end.
FIG. 5	Four-tube 2:2 directional allocation on a common carrier.
FIG. 6	Four-tube 3:1 directional allocation for an input-dominant AI or high-performance-computing workload.
FIG. 7	Four-tube 1:3 directional allocation for a distributed sensing or result-collection workload.
FIG. 8	Detail of a multifunction seam strip providing copper conductors, alignment, retention, and thermal conduction.
FIG. 9	Example arcuate segment stack including dielectric build-up layers, RDL, optical coupler, photodetector or modulator, through-substrate conductor, and chiplet bond pad.
FIG. 10	Example fabrication process from planar segment panel to curved segment and assembled tubular package substrate.
FIG. 11	Hybrid integration with a planar organic carrier and local semiconductor bridges.
FIG. 12	Example signal-flow diagram for three downlink tubes and one uplink tube.

19. INVENTOR DISCLOSURE STATEMENT

I hereby declare that I am the inventor of the subject matter described in this disclosure. This draft is intended to document the architecture and contemplated variants before proof-of-concept collaboration and before public disclosure. The final filing should be reviewed for completeness, inventorship, drawing support, and consistency with any additional embodiments developed before submission.

Inventor Signature: _____

Printed Name: River Chen

Date: June 2026

20. DRAFTING NOTES FOR FINAL FILING

- Prepare formal line drawings before filing. The drawings should show both a single-tube architecture and at least the 2:2 and 3:1 four-tube directional embodiments.
- Avoid presenting projected power, cooling, skew, energy-per-bit, or heat-flux values as measured results unless supported by simulation or prototype evidence. Preserve the distinction between total heat generation, effective heat-transfer area, and heat-flux density.
- Confirm that all intended material alternatives, optical-path alternatives, seam structures, and chiplet attachment methods are expressly disclosed before filing.
- Consider filing an additional provisional application if a PoC partner later develops materially new optical, mechanical, or manufacturing embodiments not supported by this disclosure.